NETWORK EFFECTS

2nd Edition
updated and expanded



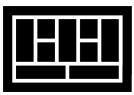
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SUPER GUIDE:

NETWORK EFFECTS

BY DANIEL PEREIRA



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Daniel Pereira
The Business Model Analyst Ottawa, ON, Canada
businessmodelanalyst.com

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INTRODUCTION

Schools frequently teach about the concepts of demand and supply and that when pricing your products, services, or platform, you should consider the unique value of your products and consumers' willingness to pay for them. Still, the most essential factor that could impact most companies' pricing strategies is frequently overlooked.

We all know how big Meta/Facebook, Uber, Alphabet/Google, Apple, and Amazon are in the digital space; they are even considered "digital giants," making billions of dollars annually. Of course, each one is very different in many ways, but there is one property that defines them all and lies behind their success. This is known as "network effects." All the companies mentioned above harness and fully utilize network effects in some way.

Even though the idea of a network effect may seem complicated at first, it is actually pretty simple. When platform builders understand network effects, they can plan how to quickly develop and grow a platform. The reason why growth marketing techniques are used is because of the network effect.

It is important to note that not all businesses can develop network effects, and not all should. There are many different approaches and strategies, and they depend on the type of product, service, market, and industry. However, the idea of network effects, in which the network becomes more valuable to users as more people use it, has been the basis for all the most successful businesses and products of the internet era. If you want to start a business, this effect could help you.

So, what are network effects?

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WHAT ARE NETWORK EFFECTS

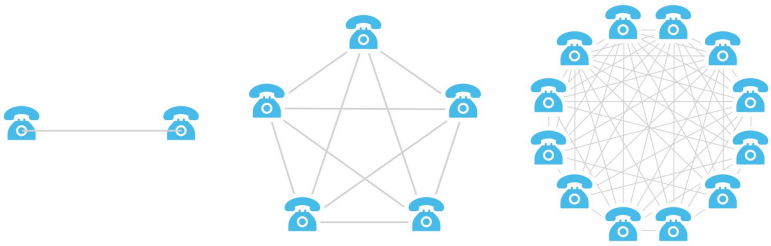
Network effects are the extra benefits that a user gets when someone else joins the network. Network effects can be found in any network, including old-school landlines, the internet, and platforms such as Instagram and Facebook.

For example, Instagram is only valid if other people (users) also have an account there. If only one person owns an account, the value of the Instagram network is zero because they cannot do anything with it. If a second person creates an account, the first person can send a message to the second person, and if this is beneficial to both, the network has some value.

According to the online course Economics for Managers, the term "network effect" refers to any circumstance where the value of a platform, service, or good is determined by the number of buyers, sellers, or users who leverage it. As the number of buyers, sellers, or users increases, so does the network effect and the value created by an offering.

"In other words, as the number of buyers or sellers for the business grows, so does the willingness to pay for a buyer," says Harvard Business School Professor Bharat Anand in the course. In a nutshell, the network effect is a situation in which the value of a good or service goes up as more people use it or take part in it.

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WHAT AREN'T NETWORK EFFECTS

People often mix up network effects with each other and with related but different business ideas. To fully grasp an idea, it is necessary first to capture what it is not.

EFFECTS ARE NOT VITALITY

The more individuals sign up for a product, the quicker it will expand – up to a certain threshold. A product is considered to be viral if the adoption rate grows with each additional product user. They are sometimes mistaken for one another because both network effects and virality contribute to an increase in the growth of a business and the value of the firm as more users join. Nevertheless, as we will see in the following examples, which are two entirely separate ideas.

Some items have the characteristics of virality but do not have network effects. Email and other cross-platform communication solutions are a good illustration of this notion. An essential element of these solutions is that they either are compatible across networks (as is the case with Hotmail) or exploit an underlying network to facilitate both the spread of viral infections and the delivery of value propositions.

Network effects and virality are not interchangeable terms. Something can go viral, like a TikTok sound or a new and popular product. This means that as more people hear about it and share it, each of those people also shares it, and the

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rate at which people discover this new thing increases exponentially; therefore, the rate at which it is considered to be viral increases exponentially. Nevertheless, the fact that I watched a TikTok video does not in any way improve the quality of your experience of watching the same film; hence, this is an example of virality rather than network effects.

Others show network effects without showing any signs of being infectious. It's possible that products with indirect network effects, like marketplaces, won't spread like wildfire. In these situations, network effects are caused by the combination of the two sides. Each side can be brought on through a viral effect by offering an incentive. Still, it's hard to use the indirect network effect to get users on one side to join through invitations or interactions from users on the other side. That is to say, encouraging users to join the conversation through interactions with other users is extremely challenging.

NETWORK EFFECTS ARE NOT ECONOMIES OF SCALE

It is not a secret that Apple sells a lot of iPhones and gains a lot from them, but this does not suggest that the company has network effects; rather, it only means that the company benefits from economies of scale. The term "Network Effect" refers to the fact that people want to buy it so that they can use iMessage to talk to their other friends. So, what is the economics of scale?

The term "economies of scale" refers to situations where there is a large enough amount of production to save a lot of money. Because of this, the biggest player in the industry can keep making the most money possible.

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Network effects are not the same as economies of scale because, unlike economies of scale, they provide more value for a given marginal rise in cost. The price goes up when networks get bigger, but the product's overall value skyrockets at an even faster rate.

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NETWORK EFFECT VS. NETWORK EXTERNALITY

It's important to note the differences between network effects and externalities, despite their similarities. The term "network externality" is used in economics to describe how the demand for a product is affected by the desire of others to purchase the same thing. That is to say, when people see that others around them are purchasing a certain thing, it tends to sway their own decisions to do the same.

For example, you might decide that a book's content is worth reading if a lot of people have shared it on their platforms. You decide to give it a shot because, surely, so many people can't be incorrect. Customers' spending habits are impacted by more than just price tags and discounts, though. When figuring out if an item of clothing will be trendy in the future, the main thing to look at is how quickly people buy and sell similar items.

Assuming there are positive network externalities, a network effect may occur. To the extent that many of your friends already use Facebook, that may entice you to sign up as well. After signing up, if you contribute something of value to the community in the form of content, and as a result, more people start to actively participate because they are having a good time, a network effect will be created.

The term "network effect" refers to positive network externalities, which happen when the value of a product or service goes up as the number of people who use it grows.

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When the product's value goes down as the number of users goes up, this is called a negative network externality. Congestion is another name for this phenomenon.

When a certain number of people sign up for a service or buy a product, the network effects start to matter. The term "critical mass" refers to a specific level of subscriptions. Once a certain threshold is crossed, the benefits of using the product or service outweigh the cost.

After a critical mass of consumers has bought the product, more will do so since its value has increased relative to its cost. For such firms to succeed, reaching a critical mass is essential. Companies use different kinds of advertising, like giving away free products and offering discounts, to get new customers.

Businesses can attract customers by lowering prices or asking them to invite their friends to join. It is prudent to create a system that is valuable to early adopters even in the absence of network effects. The more people who utilize the system, the more valuable it becomes, and the more people will sign up for the service.

Once a service reaches a certain level of popularity, the rate of growth in its subscriber base often levels off. After a given amount of time, the networks get overloaded. Congestion occurs when there is an excessive demand for something. Let's use mobile networks as an example. After a certain threshold, adding more users actually hurts the experience for everyone. When the network is too busy, it causes interruptions in service and a lack of connectivity.

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HISTORY OF THE NETWORK EFFECT

Network effects was one of the main arguments that Theodore Vail, the first president of Bell Telephone after the patent ran out, used to try to get a monopoly on telephone services in the US. In 1908, when he wrote about the idea in Bell's annual report, there were more than 4,000 local and local phone networks, the majority of which were absorbed by the national Bell System.

Metcalfe's law is a way of talking about network effects that he made popular. Metcalfe was a co-founder of 3Com and one of the people who came up with the idea for Ethernet. Metcalfe used the idea that customers needed a certain number of Ethernet cards to get the most out of their network to help sell the product. Metcalfe said that the reason networking cards were sold was that the cost of the network went up with the number of cards installed, but the value of the network went up with the square of the number of users. Algebraically, this was written as having a cost of N and a value of N2. Even though the exact numbers were never clear, the idea was that customers could share expensive resources like disk drives and printers, send e-mail, and eventually connect to the Internet.

Between 1985 and 1995, economists Michael L. Katz, Carl Shapiro, Joseph Farrell, and Garth Saloner made a lot of progress in the theory of the network effect. At BlackHat and Defcon in 2009, author and high-tech entrepreneur Rod Beckstrom gave a mathematical model for describing

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networks that are in a state of a positive network effect. He also gave an economic model for describing the "inverse network effect."

Because the network effect often involves positive feedback, system dynamics can be used as a modeling method to explain it. Word of mouth and the Bass model of diffusion could also be used. Between 2000 and 2003, researchers Geoffrey G. Parker, Marshall Van Alstyne, Jean-Charles Rochet, and Jean Tirole independently developed the two-sided market literature, which shows how network externalities that affect more than one group can lead to free pricing for one of those groups. This was the next big step forward.

Despite the fact that the number of sources is decreasing, the functionality of new services, products, and apps such as Spotify, Dropbox, and messaging platforms (Messenger, WhatsApp, and Snapchat) are improving. Another important finding was that the "infant mortality" rate of websites went up a lot, with the top sites in each functional niche protecting their territory more than ever.

Metcalfe's law drove early network effect research. Metcalfe's law says that the value of a telecom network is equal to the square of the number of connected customers. In the 1990s, Joseph Farrell, Michael L., Carl Shapiro, and Garth Saloner created the network effect idea. These researchers invented the concept of using the telephone.

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WHY ARE NETWORK EFFECTS IMPORTANT?

James Currier, in his book The NFX Bible, notes that research shows that network effects are responsible for 70 percent of the value added to technology since 1994. Platforms have always been understood to have network effects.

If you want to incorporate network effects into your products, you must first recognize that not all network effects are created equal. It has been hypothesized that the most popular vendors will do best due to the network effects they generate. Businesses can benefit from network effects by fully comprehending how they function in their marketplaces.

According to the laws of network effects, consumers are willing to pay more for widely adopted products. Economics for Managers states that the fundamental principles of network effects predict that the company, website, or platform with the largest market share will be more successful in the long term.

NETWORK EFFECTS AND PRICING

Typically, businesses will set their prices as high as they can without driving away customers to achieve the highest possible profit margin. However, when network effects influence a market, market share is more important than profits, especially in the beginning.

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Before putting a price on your product, service, or platform, you must determine if network effects exist in your market. Why? For the simple reason that in markets where network effects are most pervasive, the logic behind a typical pricing strategy turns on its head.

The emergence of YouTube as a video platform is an excellent example of this concept in practice. When YouTube launched in 2005, it was a free platform to be used by all. Due to the free package, the platform became more popular, capturing a greater market share. It wasn't until 2014 that YouTube introduced YouTube premium, an ad-free version of YouTube, to fully monetize the platform.

This is because the number of active users directly affects the willingness to pay potential new consumers. You may take full advantage of network effects and maximize the adoption of your offering by boosting your market share early on, giving you the leverage to charge more once the time is right. Consequently, many businesses initially provide their wares at low prices or even for free.

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FEEDBACK LOOP OF NETWORK EFFECTS

The term feedback loop is employed in many areas of knowledge, from biology and physics to business. A feedback is the information received as a reaction to a product. This information can be used to modify the original product, to create a better one. A feedback loop is, therefore, the process by which the feedback and outputs of the customers are circled back and used as inputs.

This interdependent cycle between producers and consumers is known as a “positive feedback loop”. In a positive feedback loop, two events are mutually reinforcing: a change in one can cause a bigger final increase in the other, and so on. This is especially useful for multisided platforms.

In spite of that, it is important to highlight that not every growth on platforms leads to a higher quality service. Because the growth will attract both high and low quality users. That is why it is imperative that a high standard for access is maintained.

Uber is a classic two-sided marketplace which is benefited by the feedback loop. More passengers attract more drivers and vice-versa. More drivers mean higher availability and lower waiting time, which will be perceived as value by the passengers’ side. More passengers, on the other hand, mean lower downtime and higher prices (due to Uber’s surge pricing).

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DIRECT VS INDIRECT NETWORK EFFECTS

Network effects are broken into two categories, namely: the indirect network and direct network.

Direct network effects happen when the number of users of a product, service, or platform goes up, which makes the value of the product, service, or platform go up and the network itself grows.

Direct network effects are the most important for social media platforms because the service's value increases as more people use it. The more people who use a service, the more valuable it is. An example is the use of telephones; It is only helpful if the people you need to contact also have telephones. The more people have phones, the more valuable it is for you to have one.

Apple is another excellent example of direct network effects. iMessages sent from an iPhone to another Apple device get special treatment, which has helped the company grow its lead in the market.

When a platform or service depends on two or more user groups, like producers and consumers, buyers and sellers, or users and developers, this is called an indirect effect. As more members of one group sign up for the platform, the platform's value increases for the other group. In order for indirect network effects to happen, there must be at least two user groups.

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Taking Lyft as an example, as more riders (consumers) join the platform, it becomes more useful and valuable to drivers (producers) because it gives them more chances to make money. This is also true in reverse. As more drivers join the network, riders can get rides faster and from more places. This makes the network more useful.

Direct network effects don't work well for platform businesses because two or more user groups exchange value. There are two user groups on most platforms: producers and consumers. The more people who use a network, the more valuable it is to those who make things for that network, and vice versa. When the number of people using a product, service, or platform grows, the value of that offering rises directly as a result of the expansion of the network.

The type of user that enters a network is an important differentiating factor between indirect and direct network effects. For continuity, let's say that a new driver joins Lyft.

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PROPERTIES OF NETWORKS

NODES AND LINKS

Consumers, devices, customers, buyers, sellers, brokers, etc., are all nodes in a network. In the same network, different kinds of nodes can play very different roles. Networks are a combination of nodes and links.

Nodes can have different levels of impact, influence, power, and value, even though they are all part of the same network.

Central nodes are network nodes with many connections to other nodes and are usually more valuable. Most of the time, marginal nodes have less value because they have fewer links to other nodes. However, there are exceptions if marginal nodes are connected to a few powerful nodes. From network to network, it is tough to figure out how much a node is worth.

The number of nodes in a network is another way to measure its size. Size alone doesn't tell you how valuable a network is because the amount of activity in a network can vary.

In a network, links are the ways that nodes or groups of nodes can connect to each other. In a network, the links between the nodes are not all the same. The direction of links can be different (see below).

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Links have different strengths, which depend on how long they last, how close they are, and how often they talk to each other. For example, the link between you and your best friend on Facebook Messenger is much stronger than the link between you and someone you haven't talked to since high school. However, they both count as links in the Facebook Messenger network.

NETWORK DENSITY

Density is measured by the number of linkages relative to the number of nodes in a network. In general, the strength of network effects increases as network density rises. When two nodes are linked together, their connections are strengthened by the links between them.

Evenly distributed density is rare in a network. In some parts of a network, the population density may be significantly higher than in others. Understanding how nodes link with one another is helpful when designing products that increase network density.

Find the densest, active area of your network, or the "White-hot center," and design your offering around that area.

DIRECTIONALITY

Network science's graph theory allows two nodes to link in either direction. A graph is directed if its nodes are connected. Directed connections indicate one node points to another, but not vice versa. For example, celebrities and politicians have many Twitter followers they don't follow back. The relationships are not mutual, unlike Facebook Messenger or WhatsApp, which need the other person to respond. It has to be mutual: Facebook Messenger lets you send and receive messages. Facebook Messenger and WhatsApp are

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instances of networks with undirected connections.

Networks usually have directed and undirected connections. Drawing a map of your network's links helps you build better products and prioritize features. Interactions between network nodes determine link direction.

Central nodes with one-to-many links can broadcast to marginal nodes but seldom interact back, like the relationship between a celebrity and their followers on Instagram.

Central nodes can also occur in one-to-one networks like Facebook (although they ultimately added a one-to-many "follow" option), where some users have many friends and others few. However, one-to-many networks allow for more inequality.

ONE-TO-ONE VS ONE-TO-MANY

Network nodes can be one-to-one or one-to-many. One-to-many connections are unidirectional, directed links. One-on-one partnerships are mutual. They're aimless. Two-way interaction.

In asymmetric-follow personal networks like Twitter, Instagram, and YouTube, core nodes have many followers (inbound directed connections), whereas periphery nodes have few followers. Central nodes provide material, whereas periphery nodes watch.

Central nodes with one-to-many links can broadcast to marginal nodes but seldom interact back (think of the relationship between a celebrity and their followers on Instagram or a TV network and their viewers).

Central nodes can also happen in one-to-one networks like

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Facebook, where some users have a lot of friends, and others don't. Facebook eventually added a "follow" option for one-to-many networks. However, one-to-many networks allow for more inequality.

CLUSTERING

Real-world networks have uneven node distribution. They create tighter local groups than the network as a whole. A bridge connects two unconnected clusters via a single link.

In Slack and Facebook Messenger, people build active subgroups. Consider your private use of those services, for instance. Popular Twitter and YouTube users also congregate.

As a network grows, Reed's Law, which is explained below, predicts exponential value rises for networks with greater clustering coefficients. A high-clustering network expands exponentially, while a low-clustering network grows slowly. Not all networks cluster; however, you can increase your network's clustering coefficient.

CRITICAL MASS

A network reaches critical mass when its value exceeds that of the product and rival products. Network type determines the timing.

Telephones are early critical mass networks. In 1908, the AT&T chairman stated, "A phone without a line is not even a toy or scientific equipment. It is useless." A telephone network with two users is worth more than a single product, since a phone without connections is nothing.

Even without apps or applications, Windows is valuable. After the network of users and developers grows broad, the value

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of all third-party programs and interoperability with other users exceeds the value of Microsoft programs alone.

Most products with network effects need critical mass to fully exploit their defensibility. The product is vulnerable and may not be helpful until the network reaches critical mass. The issue for such goods is to generate enough initial value to incentivize early adopters before the network effects benefit kicks in.

IRREGULARITY

In practice, networks rarely display a single characteristic. It's exactly how they look in blueprints. They exist in groups, with both active and inactive areas. These replicate the irregularities of complex systems in the actual world.

You need to notice these differences, find the "white-hot center" in the network, and focus on it first to create a network effect before turning your attention to the larger network.

REAL IDENTITY VS PSEUDONYMITY VS ANONYMITY

Creating a public profile is a common need of network effect enterprises. In general, network effects are more robust in systems where each node's profile is associated with its true identities, such as a real or corporate name.

Real identity is essential in two-sided marketplaces and platform firms with a network impact, since trust and reputation are key to the fluidity of transactions.

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However, there are use cases where anonymity on the network is essential, such as in cryptography or government spying. However, such networks typically fail rapidly for three reasons:

1. When the quality of the data being transferred between nodes decreases as anonymous free riders drag the network down;

2. When anonymity is compromised, and trust in the system is lost;

3. When the anonymous network is no longer acceptable to the rest of the world, different authorities (like governments) step in.

And to be clear, while the conversations sent through Telegram are encrypted, the service is valued by its users because it enables them to communicate with known individuals rather than complete strangers. Confidentiality is not the same as anonymity.

ASYMMETRY

Typically, this phrase is used in reference to markets of various sizes (1-, 2-, 3-, or N-sided). Almost always, one side of a market or a certain variety of nodes is more difficult to acquire than the other.

The market's buyers, or demand side, can be challenging in particular circumstances. If you can bring in customers willing to pay, sellers will likely appear without much further work on your part. A "demand-side marketplace" describes this type of market.

Companies like Fiverr and Upwork use demand discovery and supply-side organic growth to thrive. Another illustration

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is Lending Club. They put their efforts into locating the demand (the borrowers), and the lenders respond by making the loans.

Sometimes the supply side is the more challenging one, but once that is solid, the demand side customers are drawn in naturally. This type of market is known as the "supply side" market.

For instance, Uber and Lyft spend the majority of their paid acquisition expenditure on driver acquisition — the supply side. Both sides can be equally challenging to entice in some markets, especially if you lack good fortune or sound judgment in your chosen field. That makes things very hard.

Asymmetry within a side or type of node is another kind of asymmetry in a marketplace. In other words, supply and demand are not always equal. Some nodes may be worth as much as a thousand times as much as others to have on your network.

Examine the market for these imbalances and decide which forms of demand, or supply you want to focus on bringing in first, second, and third. Then, devote your efforts to devising strategies that would allow you to successfully breach the most valuable target first.

HOMOGENEOUS VS. HETEROGENEOUS NETWORKS

Homogeneous networks are those in which each and every node serves the same purpose. Each user is essentially identical to every other in terms of their primary purpose. For example, in a landline phone network, each phone serves basically the same function as any other, and people tend to

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purchase telephones for the same reasons. Generally speaking, telecommunications networks tend to be quite standardized.

In a heterogeneous network, the nodes are divided into two or more distinct groups based on their purpose. Planners for events use the HoneyBook marketplace platform in a way that's distinct from photographers and florists. It's important to note that eBay buyers and sellers have very different motivations for participating on the network.

ASYMPTOTIC NETWORK EFFECTS

Asymptotic network effects are those that have declining returns. As a product's utilization grows, so does its value to each consumer. However, network effects can begin to fade beyond a certain point in the network's expansion. After a certain size, growth in an asymptotic network no longer benefits the existing users.

This is shown by the fact that Lyft riders don't benefit much from more drivers after they've been waiting for about 4 minutes. As the value growth on the demand side gets closer to zero, the value of more supply "asymptotes."

Asymptotic data network effects, for example. After a certain size, the method no longer improves appreciably as the dataset grows. This feature affects the majority of data network impacts. Businesses like Waze are the best at avoiding this because their service requires real-time data that must be constantly updated by hundreds, if not tens of thousands, of nodes in order to be minimally useful.

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SAME-SIDE NETWORK EFFECTS

These are direct network effects that happen on the same side of a network with more than a two-sided or N-sided network. Same-side network effects are the changes in value for users on the same side that happen when more users join that side.

Lyft, for example, has negative same-side network effects or a congestion impact. This is because a higher number of riders at any given time entails a higher price point or a longer wait time per ride. The same holds true for drivers: more Lyft drivers equals more competition for other drivers.

However, same-side network effects can be beneficial. This is especially true for Windows users, who profit from adding additional Windows users due to file compatibility. Two Windows users can easily share files with one another, and the number of individuals with whom you can share files grows as more people start to use the same platform.

CROSS-SIDE NETWORK EFFECTS

Cross-side network effects are direct network effects caused by complementary commodities or services in a multi-sided network. Cross-side network effects, which are different from indirect network effects, are when users on one side of a network get a direct boost in value caused by the addition of users on the other side.

To use Lyft as an example, there are significant positive cross-side network effects because each additional driver adds direct value to all passengers (up to a degree) and vice versa.

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INDIRECT NETWORK EFFECTS

When one type of node benefits another type of node directly but not the other nodes of its own type, we say that there is an indirect network impact. By increasing the benefits to complementing users on the other side of the network, nodes on the same side of the network indirectly benefit each other.

In a 2-sided market like eBay, for instance, the addition of a new seller does not help the existing vendors in any obvious way. In reality, additional sellers on eBay only increase the level of competition already present. However, because a larger selection of products makes the marketplace more appealing to shoppers as a whole, the presence of more sellers ultimately benefits all sellers. Adding a new seller dramatically increases the network's indirect value to existing sellers.

OS platforms like Microsoft Windows provide a further compelling illustration of indirect network effects. The addition of new Windows developers does not help the existing ones. However, the platform's user base will expand as more products are developed for Windows. In addition, more people using Windows means more people who could potentially buy the software.

NEGATIVE NETWORK EFFECTS

Negative network effects can occur as more people use the network or the network grows in size. There are two types of unfavorable network effects: network congestion (higher use) and network pollution (increased size).

Congestion in a network is most easily understood with the help of the ubiquitous phenomenon of rush hour traffic. Each additional vehicle on the road during rush hour reduces the

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efficiency of the road system in a city. Congestion is also a problem in the telecommunications network. However, it is less common than on the internet network.

However, internet networks are more susceptible to contamination than offline ones. The more people you add to your Facebook "social graph," or network of friends and contacts, the more your News Feed will be cluttered with posts from people you don't know very well or find very interesting, such as work or family connections.

Both positive and negative network effects can exist in a network. Twitter and Facebook are great examples because they display updates from many people in your network, but if there are too many updates in your feed, it can get cluttered.

Founders need to be aware of this so that they can design their products in a way that maximizes positive network effects while minimizing the impact of any negative ones.

LAWS OF NETWORK EFFECTS

Various early pioneers in the field of networks have tried to model the value-adding effects of network expansion over the years. What this means is that they made an effort to define the impact of network effects. Each subsequent law found that previous laws had grossly undervalued networks and network growth.

Contrary to the scientifically established rule of gravity, these laws are not actual laws. They are only mathematical notions that explain the use of networks and the connections between various types of networks. Since the word "law" sounds so official, it has been applied to these rules. These "rules" are not mutually exclusive, and multiple can apply to the same network at the same time.

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The basic fundamental rules presented by Sarnoff, Metcalfe, and Reed define the growth of network value as a function of network size and structure. The laws need a quick examination.

Sarnoff's law

The man who oversaw the Radio Corporation of America (the company that eventually became NBC) from its inception in 1919 until his retirement in 1970 was David Sarnoff, a true radio and television legend. In those years, this network was among the largest worldwide. Sarnoff noticed that the network's worth seemed to grow in direct correlation with its size, or proportional to N, where N is the total number of users.

While Sarnoff's description of the utility of a network was spot-on for broadcast networks in which a small number of nodes disseminate information to a large number of nodes at the periphery, it turned out to be an underestimate for other types of networks (a radio or television audience).

Metcalfe's law

According to Metcalfe's Law, a communication network's worth increases as its user base expands. In the early 1980s, Robert Metcalfe, a co-creator of the Ethernet standard, is credited with developing this idea.

Since the total number of connections in a network grows at a pace proportional to N2, Metcalfe's Law appears to hold true. The term "network" was developed to represent communication networks like Ethernet, fax, and phone systems, but with the Internet, it has expanded to encompass social networks and online marketplaces.

Reed's law

In 1999, MIT professor David P. Reed published what has become known as Reed's Law. Reed argued that

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"group-forming networks" allow cluster formation (as described above) and scale value even faster than other networks. According to Reed, the value of networks that cluster together grows at a rate of 2N, where N represents the number of nodes in the cluster.

Because there are an infinite number of groups that can form within a network that "supports easy group communication," Reed proposed a formula of 2N rather than N2 so that the density of the network is not just a function of the number of nodes but also of the number of connections between them (N2). In fact, it's proportional to both the number of nodes and the number of possible sub-groupings or clusters, and it expands exponentially with the number of users in the network.

As Reed's Law predicts, the value of online networks will increase at a rate far higher than that predicted by Metcalfe's or Sarnoff's laws because of the widespread availability of tools that facilitate the establishment of clusters inside the networks.

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TERMS OF NETWORKS

- **Network:** is a group of interconnected nodes, which may be people, such as in social networks, or things, such as in telephones or computers networks.
- **Marketplace:** is a network where transactions take place between two or more sides with distinct customer segments on each. It aims for supply and demand to get attracted to the same place.
- **Platform:** is a multi-sided network where users, developers and the platform itself create an environment which increases value for every and each of the groups. It may also be programmed, customized and extended by outside users.

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THE DYNAMICS OF NETWORK EFFECTS

The most successful companies and products of the internet age have all been based on the idea of "network effects," which says that as more people use a network, it becomes more valuable to them. This is true for companies like Amazon and Google, as well as some open-Source projects and cryptocurrencies like Wikipedia. At its core, the theory of network effects says that platforms and products with network effects get better as they get bigger. This is true not only for the value they provide to users, but also because they get more resources to improve their product, which makes the "flywheel" stronger.

But lately, theory and reality seem to be getting further apart.

We're no longer seeing markets where the winner takes all. Instead, we're seeing companies with network effects, like messaging apps and sneaker marketplaces, split markets among many players. Even companies that seem to have won the market at first and built a deep moat, like dating apps and trading platforms, are having trouble keeping their position in the face of copycats and new competitors. Just look at what Instagram Stories is doing to Snap. It recently surpassed it even among the teens who use Snap the most.

Does this mean that network effects as we know them will no longer exist? No, but they're more active than they've ever been.

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We know that not all network effects are the same, and we also know that they don't change in the same way. There are different kinds of network effects for each product, and they change and grow in different ways over time. If anything, businesses with network effects change faster than they ever have. So, how can entrepreneurs and founders survive in an era when network effects appear to be fading? The trick is to know what your network effects look like now and to predict how they will change over time. To do this, you'll need to know how three parts of your business could change in the future:

1. Your value proposition
2. Your users and inventory
3. Your competitive ecosystem

VALUE PROPOSITION

As a company or product grows, its network effects sometimes keep going in a straight line or a line with increasing returns. They could reach an asymptote, hit an inflection point, or even turn around. The most important thing for founders to know is what value proposition drives their network effects, how strong or weak they are, and how they will change, especially as they iterate their way to new value propositions and more layers of product-market fit.

Let's look at a couple of examples:

1. Social borrowing:
Sometimes, changes in network effects aren't caused by things like wait times, but rather by things like absolute liquidity (like the number of people in a network). Take the case study of Frank. This website lets people borrow money from and lend money to friends and family. Early on, having

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more friends in a Frank group meant that there was more demand and more money in the group. This made it more appealing for people to join those groups. But when there were more than seven people in a group, they were less likely to lend or borrow. It turns out that people only trust about seven friends or family members that much. In this case, the network effects went from being good to bad when an individual's network grew bigger than the value proposition. This pattern has also been seen with several other products that are used by many people.

2. Ride-sharing:
In any area, the number of drivers and the number of people who want to ride reinforce each other. More drivers mean shorter wait times, which means more people want to ride, which makes more drivers want to drive, and so on. But that value proposition has a catch: Once the 5-minute wait time is reached in a certain area, riders don't care whether there are more drivers in the network. And once multiple platforms reach a certain level of relative liquidity (for example, when there are enough drivers to meet riders' 5-minute wait time), riders care less about which platform they use. In those markets, they'll have to compete in other ways, like brand or reputation, price, user experience, loyalty programs, and more. This means that the value proposition needs to change.

3. Networks of people:
As Facebook changed from where friends could share status updates to where news and content could be found, the network effects got weaker. People didn't feel as comfortable sharing personal content when they had too many friends or followers, so they shared more news and public content. So, the value proposition and network effects also changed from social networking to social media. Importantly, adding more nodes was less useful for finding media than it used to be in the early days of a pure social network. So, the network effect curve hit an inflection point when the value proposition

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changed. Some features, like curation algorithms and timelines, helped put off this tipping point by letting users add more friends while still making the content feel relevant. But eventually, the content mix, the value proposition, and the network effects all changed.

The point of giving these examples is to show how network effects work and how they change over time. If you don't pay attention to these things, you might think that network effects no longer exist in a certain business when finding new ways to unlock value.

USERS AND INVENTORY

Your product or platform's current users and inventory and the types of users and items you're adding are keys to understanding and predicting the future of your network effects.

Commoditized vs Differentiated supply

When predicting network effects, especially for platforms and marketplaces with two sides, it's important to know if users and inventory are similar or different. When it comes to ride-sharing, the customer (the rider) only cares a little about the underlying service provider or inventory because they see the supply (drivers, cars, and transportation) as interchangeable and, therefore, a commodity. Once a platform has reached a base level of liquidity, its network effects are more likely to reach an asymptote if it has a lot of inventory that can be treated as a commodity. For the ride-sharing category, moving into related businesses (as Lyft did with its healthcare initiative or Uber with food delivery) lets companies offer different, but still interchangeable products. This could make the network effect stronger.

Network effects are stronger and last longer when there are

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more products on a platform or market. This is because they offer a variety of products that meet the different needs of customers (while maintaining just enough substitutability across that inventory as well). For example, Airbnb can show users every type of lodging in Los Angeles that costs between $225 and $325 per night, which overlaps with someone else's search for a place that costs between $150 and $250 and has a balcony and a hot tub. Because of this, the platform is more valuable on both sides of the market than a site that only shows a set of standard and executive rooms that are all the same.

The network effects stay strong not only because it reaches a base level of liquidity across all these different types of inventory (making them valuable to more users) but also because it continues to see increasing returns with new supply.

But the more unique the items are, the better the platform has to be at curation and matching. This makes it easier to defend the platform as a whole and keeps the network effects curve strongly over time.

Types of incremental users

Users and inventory are treated as goods, but that doesn't mean that everyone in a network is the same. Some things are worth more or less than others. For example, a popular restaurant close to many OpenTable users is more valuable to the network than a bad restaurant in the middle of nowhere.

When you plan your network effects and, more importantly, your growth strategy for getting more users and keeping them interested, you should consider the new users you're likely to get. Are they "contaminants" in the network, "neutrals," or "contributors"? When a troll joins a social network and turns people off, they are a pollutant that takes away value. Adding a lurker does not affect the network

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because that person doesn't add to or take away from its value. Adding a great content producer to the network adds a huge amount of value.

So, it's important to make sure to reward the users you want and punish the ones you don't. This is why most great platforms, like Wikipedia's editors, Airbnb's reviews and onboarding, etc., invest a lot in ways to find and get rid of bad inventory and users. Unfortunately, these ways of weeding out weak contributors don't always work, and sometimes it costs a lot to find strong ones. In this case, the ratio of growth to cost matters a lot.

COMPETITION

To understand and predict network effects, you also need to know your market and who your competitors and alternatives are. Let's take a look at the following:

Network overlap
Businesses that benefit from network effects tend to be more secure as they grow, but new entrants in the market can threaten even the largest of firms. But for these kinds of businesses, it's not enough to just know who their direct competitors are. They also need to think about how their networks overlap. A rival may enter your market if they have access to resources that are comparable to yours. As they already have a similar network in place, they will find it much simpler to break into your space (take Instagram's recent push into Snapchat-like "Stories" as an example). Also, suppose a rival company has already filed a patent covering a larger portion of the same network. In that case, you should expect the same treatment.

Satisfaction of current needs
When users have to switch between different platforms to

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finish their tasks, the network effects are hurt. As an illustration, consider the job marketplace: since hiring is such a crucial part of running a business and no single platform is likely to satisfy all of a company's hiring requirements, employers are likely to post their openings on multiple hiring platforms (i.e., multi-tenant). The economics can be turned upside down when one side of a platform is multi-tenant due to the operator having to deal with additional pricing, feature, and liquidity requirements.

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TYPES OF NETWORK EFFECTS

Network effects can exist in different ways; the commonly known types are:

- Direct network effects
- Indirect network effects
- Two-sided network effects

DIRECT NETWORK EFFECTS

The value of commodities changes depending on demand. When someone places an order, the supply increases, and companies change their prices based on how much people think their goods are worth. This is the simplest form of a network effect.

Example of Direct Network Effects

- Telephone

Physical (Direct)
Physical (Direct) are network effects related to physical nodes, such as (mobile phones and computers), and physical connections. Most physical networks are a winner-takes-all monopoly market. Considerable capital investment is needed to rival a company with a physical network.

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Example of Physical Network Effects

- Telecommunication (Cable, Satellite)
- Infrastructure (Water, Electricity)
- Transportation (Road, Subway)

Protocol (Direct)

A "protocol network effect" occurs when consensus is reached on a standard method of communication or computation, and all nodes and producers can use this new standard to join the network.

Example of Protocol Network Effect

- Cryptocurrency
- Ethernet
- Fax

Personal Utility (Direct)

People rely on personal utility networks to talk to and interact with people in their private networks. When they aren't connected, it causes big problems. If people opt out, it would make their daily lives much harder and hurt their most important personal and professional relationships.

Example of Personal Utility Network Effect

- Instant Messaging (WhatsApp, Facebook Messenger)
- Communication tools (SMS, iMessage)

Personal (Direct)

The personal network effect originates from our friendly, tribal nature, which drives us to want to make relationships with others. People's friends, coworkers, and neighbors are also

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members of these networks, so it makes sense that they sign up for them and stick with them.

Example of personal Network Effect

- Social (Facebook, Instagram)
- Professional (LinkedIn, Twitter)

Market Networks (Direct)

Market networks combine the identification and communication features of a personal network with the transaction-centric, goal-driven structure of a market. In most cases, market networks begin by expanding upon an offline community of specialists.

Example of Market Networks

- Honeybook
- Angellist
- IvyMark

INDIRECT NETWORK EFFECTS

The worth of a product or service rises when more people or organizations find value in it.

Example of Indirect Network Effects

- OS platforms (Microsoft Windows, Android, Linux)

2-SIDED NETWORK EFFECTS

The 2-Sided network effect occurs when the value of a product or service rises across multiple user groups. The rise

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of online markets illustrates this trend. When there are more buyers in an online market, business owners may be more likely to open a store there.

Marketplace (2-Sided)

In a 2-sided marketplace, the network is more important than the app or website itself. You'll need a more compelling offer to pull them apart that is suitable for both the buyer and the seller.

Example of 2-Sided Marketplace Network Effect

- Airbnb
- Fiverr
- Uber
- eBay
- (Visa, Mastercard, American Express)
- Media company (Wikipedia, Meta/Facebook)

Platform (2-Sided)

A "2-sided platform" is a network effect where both sides mutually benefit. Suppliers' output and sales are inseparable from the platform they operate on. They buy it not just for the product, but also for the name.

Example of 2-Sided Platform Network Effect

- Mobile operating system (Android, iOS)
- Desktop operating system (Microsoft Windows, macOS)

Asymptotic Marketplace (2-Sided)

In a "2-sided asymptotic" marketplace, the demand side's

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value rises as supply rises. These shopping centers are more likely to have more than one tenant because more than one person could rent the space.

Example of Asymptotic Marketplace (2-sided)

- Uber
- Lyft

Data Network Effects

In the case of data network effects, the value of a product rises as more data is collected and used, and the more people who use the product, the more data they collect. Data's power in a product lies in how much it aids its consumers.

Example of Data Network Effect

- Google
- Amazon
- IMDB

Tech Performance Network Effects

The more people who utilize the network, or the more devices that are connected to the network, the more beneficial the technology becomes. Because of its technical performance, your product will have a head start on the competition. No effort is required to ensure that your lead grows rather than shrinks.

Example of Tech Performance Network Effect

- P2P (BitTorrent, Ares Galaxy, FrostWire)
- IoT

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“SOCIAL” NETWORK EFFECTS

Social network effects are the final category of network effects. The more people who utilize your product, the more value you'll be able to build for them. Language, belief, and bandwagon are the most common forms of social network impact.

Language

Language has always been an essential part of how people in a social network talk to each other. Startups can use the winner-takes-most effect in at least two ways by using the network effects of language. A founder's voice is bolstered if they are instrumental in coining a new business category and becoming recognized widely as an industry leader in that space.

Example of Language Network Effect

- Company names: Xerox, Uber, Google,
- Broad language networks: Language families, dialects, individual languages,
- Category names (Lite Beer, Yahoo! portal, Bitcoin cryptocurrency)

Belief

The greater the conviction of those who support a cause, the greater the likelihood it will succeed. Accepting a group requires a person to share the group's values. In other words, the greater the number of people who believe something, the more valuable it becomes.

Example of Belief Network Effect

- Religions

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- Ideologies
- Currencies

Bandwagon
Bandwagoning refers to peer pressure to join a group or movement to avoid feeling isolated. Apple's success depends on appealing to people's desire to fit in with the in-crowd and on getting them to embrace the trend. The trend's early adopters may lose interest if too many people start participating.

Example of Bandwagon Network Effect

- Apple
- Google
- LinkedIn

Tribal
The tribal network effect works in the alum networks of schools, military units, fraternities, sororities, accelerators, languages, regions, and faiths. Members are taught to aid others and defend the tribe's honor to increase the group's worth. As a result of members of the tribe helping one another, tribal networks tend to have more nodes. Individuals put forth an extra effort to build relationships within their tribal networks because they value the opinions and support of their fellow tribe members; this increases the density of nodes, which improves network effects.

LOCAL NETWORK EFFECTS

A strong local network can help keep user engagement high and stable. A regional network is a group of people who have come together because they share the same medium

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(interest, socio-economic class, or physical location). The network will be more resistant to outside threats as a result.

NEGATIVE NETWORK EFFECTS

More use of the network or a more extensive network can harm its value in some circumstances. If you follow a lot of Twitter friends, for instance, your News Feed may become cluttered with posts from individuals you don't care about.

Network Pollution

This type of negative network effect usually applies to social media. When a user's social graph expands above a certain threshold, posts from distant acquaintances or family members might drown out those from close friends. The "signal-to-noise ratio" on the social feed drops, causing fewer people to pay attention and less value to be gained from the network.

Bad Actors

Any kind of network is vulnerable to attack by bad actors (malicious users). The most vulnerable places are networks, markets, and platforms that let users talk to each other because scammers can more easily target people who use them to take advantage of them and the market. Bad people need to be kept out of the network so that all members can interact positively.

Examples of negative network effects

Due to adverse network effects, the value of multiple platforms has been eroded. What you've established for your forum will determine how they can take it over. For example:

Google case study

Google hasn't always had an easy time dealing with spam on its index. Google's spam and lousy writing are being fought

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by software and honest people who rate the quality of content. In the long run, this would have been detrimental to the platform.

Airbnb case study
In a two-sided platform, the participation of one type of user improves the experience for the other. Having more Airbnb hosts means that guests have more accommodations from which to choose, increasing the service's value. However, the platform loses its appeal to hosts and, thus, its value.

Tinder case study
Take the Tinder app, based on the idea that users will benefit from meeting people nearby. The dating site will quickly lose its appeal if there are no potential dates in the area. More users from all over the world may not have an impact on local network effects.

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HIDDEN NETWORK EFFECTS

"Hidden network effects" are, as the name suggests, the unintended and often unnoticed benefits that people who use a product or service get from the actions of others. Since these influences can have either positive or negative results, mastering them and using them to one's advantage can be crucial to success in today's cutthroat business environment.

Here are a few examples of networks that, at first glance, may not appear to have network effects.

SLOW NETWORKS

Slow networks are a type of hidden network effect that takes a long time to show value. They are standard in education, lending, hiring, health care, and real estate, where feedback loops are long, and users interact with the network infrequently. Slow networks can take months or even years to reveal their value.

UNFINISHED NETWORKS

Like a slow network, an unfinished network is a type of hidden network effect that has yet to reach its full potential. There are unfinished networks in many fields, like social media, online marketplaces, and online communities, where the network is still growing and has not yet reached its full potential. These networks have the potential for substantial

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network effects. However, they still need to be fully developed or have yet to reach a sufficient saturation level. In short, this means that the network's benefits haven't been fully realized yet and might not be evident immediately.

THROTTLED NETWORKS

One way to describe a "throttled network" is as one in which the network's growth or participation has been slowed down or stopped on purpose. There could be a number of reasons for this, such as government restrictions, a lack of funds, or a plan by the company to slow expansion in order to keep quality in check and ensure stability. So, there are network effects in throttled networks, but they aren't fully realized because the network isn't growing at its natural rate.

The early years of Facebook are an illustration of a throttled network. Users needed a Harvard email address at first, then edu-email addresses, and finally, the entire public to use the application. Facebook limited the network's size on purpose, but didn't limit how users could interact on the platform. This made network effects clearer. Facebook was also able to manage the platform's interactions and user quality thanks to this network throttling, but it also covered just how powerful the network effects were. This is how a throttled network functions.

LATENT NETWORKS

Latent networks, also called "come for the network, stay for the tool," are a subset of the Hidden Networks Effect. They are created by establishing a community that functions as a network, connecting people, and providing value to one another. Then, they release a product or technology that boosts or speeds up the engagement of the network.

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Once the product is out, it might be possible to determine the network's strength and potential because something needs to be analyzed or sold. Latent networks are the hardest to plan for and set up because they have the most "hidden network effects."

HIDDEN ADVANTAGES

Despite the drawbacks of requiring more time, conviction, and capital, hidden network effects have significant benefits. Businesses with hidden network effects can maintain rapid expansion even when their industry experiences slower growth. This is because when more people join and participate, the network's value rises, resulting in a positive feedback loop.

Entrepreneurs can acquire an edge over the competition and create a more valuable and long-lasting enterprise by discovering and capitalizing on hidden network effects. Additionally, the value of the network rises as more people utilize it, meaning that investors in companies with hidden network effects may experience better profits. Also, scalability and profitability are increased since, as the network expands, the services' expenses remain constant, allowing for more scalability and profitability.

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HOW THE NETWORK EFFECT WORKS

The term "network effect" refers to the phenomenon in which the value of a product increases for all users, including the existing user base, when new people join a platform. This creates a cycle of growth and value creation that makes it hard for new companies to enter the market.

Two Sources contribute to the value of the network effect: **inherent value and network value**. The inherent value of a product is the benefit the user receives from utilizing the product. In contrast, the network value is the benefit the user gets from the network using the product.

The objective of making this system function is not to concentrate on what occurs after the loop has begun, but on what happens before. Some businesses use things like a payment to sign up or refer a friend, a fee waiver, or a request for a referral to get people interested. Others rely on sustainable techniques in which the product is sufficiently valued and disruptive for early adopters, if only for some.

Consequently, the influence of network effects depends on the total number of prospective buyers and sellers in the market and the extent to which the organization can leverage its user base. Nevertheless, caution must be taken with issues such as congestion and saturation. As the scale of the network effect increases, the network itself may become overloaded. This will limit the uptake. Congestion is a result of excessive use. Before the point of congestion, each

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additional user brings value to the network. But after the congestion point, every additional user diminishes the network's usefulness. If the system is not expanded, it will crash.

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STRATEGIES FOR BUILDING NETWORK EFFECTS METRICS

To take advantage of network effects and grow your business, you need to build a successful business model that gives each group of users distinctive and differentiated value on your platform.

Here are a few strategies firms can use to benefit from network effects.

FIRST MOVER ADVANTAGE

An organization or person gains a competitive edge when they are the first to enter a market. One such benefit is a head start on creating brand awareness and attracting a customer base. It's valuable to be the first to market, since the network effect occurs when the first company has a more extensive user base, which in turn draws more people.

It's also worth noting that there are downsides to having the first-mover advantage. Research and development, marketing, and product development may cost a lot for the first mover to enter a market. There's also the possibility that the first mover won't be able to adjust to shifting market conditions or technological advancements. To capitalize on the first-mover advantage, businesses would need a clear plan and awareness of the risks.

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One company that benefited from being a first mover in its industry is eBay. After ten years of effort, Amazon and Yahoo finally gave up trying to overthrow eBay as the leading platform for online auctions. This shows how important it is to quickly expand the user base early on instead of trying to catch up later.

LOCK-IN EFFECT

The lock-in effect is when customers become so dependent on a particular product or service that it becomes difficult for them to switch to a competitor. This is evident in the Android or Apple ecosystems, where customers are locked in because they have spent time and money on the company's products and services.

This effect is especially good for businesses because it makes it harder for competitors to join. But a company can only benefit from the lock-in effect if its product or service is of high quality. This is because customers will seek alternatives and switch to a competitor if they are dissatisfied with the product quality. Therefore, to ensure the lock-in effect is effective, it is essential to deliver a service or product that reflects the quality and expectations of clients.

CREATE COMPLEMENTARY PRODUCTS AND SERVICES

Companies can also benefit from network effects by developing complementary products and services, as Facebook has demonstrated by integrating its platform with popular apps like Instagram and WhatsApp. This allows firms to leverage existing networks and extend their reach.

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By creating complementary products, businesses can boost the engagement of their comprehensive platform and attract new customers who may be tempted by one of the complementary products. This strategy might be interpreted as "if you can't beat them, join them," since organizations can leverage existing networks to acquire more customers and expand their own network.

TAKE ADVANTAGE OF PLATFORM EFFECTS

Businesses can greatly benefit from creating a "platform," which is simply a product or service that can be used as a basis for developing other products and services.

This is evident in platforms like Android, which encourage the development of apps for Android-powered devices. The App Store is another example of a digital distribution platform. The more firms that join such networks, the more valuable they become, and the more businesses are drawn to them. This results in a significant network effect as the platform's value rises.

SUBSIDIZE USER ADOPTION TO GET A CRITICAL MASS

Subsidizing user adoption is a business approach in which a corporation uses its resources to reduce the price of its product or service to attract more customers.

One such method of doing this is by offering special discounts or promotions. As more people start using the

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product or service, its efficiency will increase, and its expenses will decrease. Gaining market share in this manner can be effective, but it requires a substantial investment. This approach is also potentially risky because it does not ensure long-term success. However, if implemented well, it can provide a corporation with a huge short-term edge over its competitors.

BUILD A COMMUNITY

Building user trust and loyalty through community creation can provide a significant competitive edge for a business. Airbnb is a great example of a company that successfully brought together hosts and guests by creating a safe online marketplace for short-term rentals. As more people joined, the platform became more efficient, and as a result, it was able to offer reduced costs, which in turn attracted more customers. eBay is another company that focuses on making it easy for collectors and hobbyists with similar interests to meet and talk to each other. Because of this, they rapidly expanded their user base by word of mouth.

TAKE ADVANTAGE OF TIPPING POINTS

A product or trend reaches a "tipping point" when its popularity increases exponentially. Knowing when to take advantage of network effects can be critical for businesses. When demand for a product or service reaches a tipping point, sales go through the roof, and the business proliferates. Airbnb is an excellent example of this because it relied heavily on word-of-mouth marketing in its early days, leading to a tipping point and quick growth. Airbnb was also able to attract more customers and new users by

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undercutting the prices of its competitors.

INCREASE THE ECONOMIES OF SCALE ON BOTH THE SUPPLY AND DEMAND SIDES

To capitalize on network effects, businesses must offer customers new products and services. Using cutting-edge technology can help a business stand out from the competition by giving customers better service and features. This can boost business by bringing in new clients and keeping existing ones. The Xbox video game console from Microsoft is a fantastic illustration of this. Since there is a wide variety of games to choose from, the platform is attractive to players who want a full-fledged entertainment experience. More economies of scale help both the supply and demand sides, leading to growth.

PARTNER WITH COMPLEMENTARY BUSINESSES

Creating synergistic partnerships with other businesses is another approach to capitalizing on network effects. For example, consider Amazon's partnerships with companies like UPS and FedEx that deliver packages. Amazon was able to reach more people and improve the shopping experience for its patrons by teaming up with these companies. Amazon's platform got more efficient as more and more individuals signed up for it; as a result, they were able to offer lower prices, which in turn attracted even more customers. Although this plan may require a substantial outlay of

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resources at the outset, it has the potential to be a powerful engine of expansion and a source of sustainable competitive advantage.

STRIVE FOR OPERATIONAL EXCELLENCE

If a company wants to take advantage of network effects, it should put customer service at the top of its list of priorities. Companies like Netflix have been able to keep their customers and keep them coming back by giving them a good experience and ensuring their systems can handle growth. If businesses prioritize their customers' needs, they can start a virtuous cycle in which more business from happy customers drives further improvements in customer service, and so on.

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NETWORK EFFECTS METRICS

Are businesses prepared to realize their full potential? Network metrics are here, so the answer is yes. These potent tools enable organizations to assess the strength and impact of their network, providing a clear picture of how to maximize expansion and retention. Network effect metrics provide significant insight into the inner workings of businesses, from monitoring user engagement to identifying critical tipping points.

ACQUISITION-RELATED METRICS

Acquisition-related Metrics are a company's lifeblood. These metrics measure the effectiveness of a company's marketing and sales efforts, providing valuable insight into how buyers discover and interact with a product. Acquisition-related metrics, like the number of website visitors and the rate at which leads turn into customers, are essential for determining how a company grows. In addition to giving a snapshot of a company's performance, these indicators can help businesses find opportunities for growth and make decisions based on data.

Organic vs. paid users

Network effects are a big part of business, and the ratio of organic users to paid users is one of the most important ways to measure the success of a business with network effects. As a network grows and becomes more useful for users,

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more users should wish to join independently without requiring paid acquisition. This indicates that the proportion of organic consumers should increase over time.

For organizations with direct-side network effects, like Facebook, the organic share will increase as users invite their friends to join the platform, and the network's value rises. For two-sided marketplaces such as Airbnb or eBay, the organic share of new users increases as more sellers and buyers desire to join the network due to the possibility of profit and a variety of options.

It is important to emphasize that this does not imply that paid acquisition is terrible, as many companies, especially in new areas, spend money to gain new users. After reaching a critical mass of users, a company should reduce its reliance on paid acquisition.

But it is important to figure out the right number of people for whom the organic part should grow. For a company with local network effects, like helping people find nearby home service providers, the number of organically-grown new users should grow in each area. Organic users should increase at the atomic level of a small social network designed to be used inside a certain location, such as a school.

Sources of traffic

The connection between customer acquisition expenses and network effects can be complex and complicated. Even though it may seem obvious that the cost of getting new customers should go down as a company grows, and network effects take hold, this is not always the case. Customer acquisition costs can be affected by various factors, including competition on marketing channels, the availability of substitute items, and viral loops.

In the ride-sharing business, for example, drivers have a lot of

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choices, which makes it more expensive to find new drivers over time. On the other hand, companies like OpenTable that put all of their customers' requests on one platform may find it cheaper to get new customers over time.

It is essential to remember that network effects and virality are different things. Despite the fact that organizations with a network effect have a strong viral growth component, which might impact client acquisition costs, the two factors are different. A company may grow quickly because of viral growth, but not because of network effects. This can cause both rapid growth and a quick fall.

Time series of paid CAC
Customer acquisition cost (CAC) is a metric that shows how much it costs a business to get a new customer. CAC can show how well a company's acquisition strategy is working, so it's essential for organizations to keep a close eye on it.

However, CAC can be somewhat more complex for organizations that rely on network effects. In these kinds of businesses, the product's or service's value increases as more customers join the network. As a result, CAC might go down over time. But other things can affect CAC, like competition in the marketing channels and the availability of substitute items.

Understanding the dynamics of CAC in firms with a network impact is essential for making strategic customer acquisition decisions.

COMPETITOR-RELATED METRICS

For success in the corporate world, staying ahead of the competition is essential. But how do businesses evaluate their growth compared to their competitors? Enter

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competitor-related metrics. These indicators provide vital information about how organizations compare to the competition, enabling them to make informed decisions about how to stay ahead of the competition.
Competitor-related metrics, ranging from market share to customer retention rates, equip businesses with the means to study their competitors and make strategic decisions.

Prevalence of multi-tenanting
To stay ahead of the competition, businesses need to know if their users are also using similar services. This includes related services whose functions may not be the same. This is called "multi-tenancy," and it could be a useful tool for businesses that want to offer more services and get more users. Measuring multi-tenancy can be hard because you might have to ask users questions, look at churn or usage drops, or look for users on other platforms.

Switching or multi-homing costs
Switching costs, or how much work and money it takes for a user to leave one network and join another, are key factors in whether a product can keep its users or not. Users are less likely to abandon a network that requires a substantial investment of time and resources to join if the switching costs are high. Products with low switching costs, on the other hand, are more vulnerable to multi-tenancy, or the use of multiple networks.

Switching costs can be affected by how much time and effort it takes to join a network, how much personal information and content a user has invested in that network, and how much value a user can get from a new network right away. Measuring switching costs can be hard, but the time it takes to complete a competitor's onboarding process or the number of network connections a user has are good examples of useful metrics.

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ENGAGEMENT-RELATED METRICS

As businesses try to grow and succeed in the digital world, it's becoming more and more important to understand and measure how engaged users are. Yet, what precisely are engagement-related metrics, and how might they be used to promote company success? User engagement metrics can range from relatively straightforward measurements like page views and click-through rates to more complex measures such as time spent on site or user retention rates.

By tracking engagement-related metrics, companies can gain a deeper understanding of their customer's behavior and preferences and make informed decisions about improving the user experience and driving growth. So, whether you want to keep users coming back, make more people aware of your brand, or get more sales, you need to understand and use engagement-related metrics to reach your business goals.

User retention cohorts

User retention is an important indicator to consider when evaluating the success of a product or service. Keep in mind that the oldest user cohorts, particularly for social networks or community-based products, tend to be the "ideal" customers for a product or service. These early adopters often have better retention rates, which can skew the total numbers.

When looking at user retention metrics, it's important to look at these things and remember that the results may not always match the theory behind network effects.

Core action retention cohorts

It's not enough to count users or measure general engagement to determine a product's success. This can be done, for example, by concentrating on the product's "core

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activities," or the one precise task that best exemplifies the product's unique selling proposition and is relevant to the product's business model. By looking at the percentage of users who keep using the product after taking this important step, you can learn about the product's ability to meet its goals and how its social features affect user retention.

Dollar retention & paid user retention cohorts

For subscriptions and paid products, retention is important since it reveals how much users value the service. When it comes to monetary and subscriber retention, businesses should expect to see an improvement from the newest user cohorts. This means that new users are more likely to remain subscribers or continue paying for the product as the network grows.

An example is that as network coverage improves for Angie's List, a directory of home services, new subscriber cohorts are better maintained in terms of dollar retention and the number of users that remain subscribers due to the site's increased value.

Retention by location or geography

In the local network effect, the value of a product or service increases for customers in a particular area when more users in that area join the network. It follows that the company's efforts should be directed towards increasing the network density in each area, with the expectation that retention rates will rise as the markets mature and reach a certain threshold.

For instance, the value of Care.com, a platform that connects families with caregivers, improves for Charlotte users as more babysitters join the network. But the fact that there are more babysitters in New York City doesn't change how users feel in Charlotte. This is why customers tend to stick with older or more established markets more than they do with new ones.

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Power-user curves (aka L7 & L30 charts)

A product's "power users" are the people who provide the most value to the network simply by using it frequently. Companies use measurements like DAU/MAU (daily active users divided by monthly active users) to determine how involved these customers are. However, these measurements have their limits.

Power-user curves, which are histograms that show the total number of days users did a certain action during a certain time period, are a better way to look at engagement. Businesses can use these curves as a chart of network effect success, showing whether their product improves in value as more people use it or not. The shift of the power user curve to higher frequency engagement buckets over time indicates that the product is becoming more valuable to users.

MARKETPLACE METRICS

Market metrics — a fascinating new field! Whether you're just a seasoned pro or an entrepreneur, you must have a firm grasp of the supply and demand, liquidity, and conversion rates that matter most to marketplace firms.

Match rate

The "match rate" is a way to measure how well a marketplace brings together buyers and sellers. This parameter should be monitored since it reflects the marketplace's ability to match supply and demand. In the ride-sharing sector, for instance, the match rate is calculated as the ratio of the amount of time a driver spends with a passenger to the amount of time the driver spends alone.

In job marketplaces, the match rate is the percentage of employers who can fill their posted jobs minus the percentage of people who can find jobs. Also, measuring

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"zeroes," or failed matches, can be a key part of figuring out how well the marketplace works. If the match rate is low, users may be more likely to go to other marketplaces. The marketplace operator needs to figure out why matches don't happen and take steps to remove or reduce these barriers.

Market depth

"Offer depth" or "market depth" means that a market can support a large number of buyers and sellers without changing the prices. In consumer markets, it is important to measure market depth because it has a direct effect on how the user feels. For example, a user is more likely to find what they are looking for if they open a marketplace app and see a lot of listings. For a market with similar products, market depth affects how easy it is to use. If there are a lot of items nearby, it will be easier for the user to find what they need. But in heterogeneous supply marketplaces, where each item is unique and could add value to the user, there is no limit to how deep the market can go. The goal of any marketplace is to make it easy for people to find each other and do business with them. If there isn't enough depth in the market, it could lead to lousy network effects, where too much supply makes it hard to find things and can lower conversion rates.

Time to find a match

This metric is known as "inventory turnover" or "days to turn" and reveals how quickly the marketplace can arrange matches. Inventory turnover and days to turn are measures of the rate at which goods and services are sold or matched on a marketplace. Inventory turnover is a standard metric for online marketplaces that monitors the rate at which things are sold. Days to turn is more beneficial for traditional marketplaces where user matching happens when they opt in. It measures the amount of time required for a listing to be matched or a product to be sold.

These measurements are essential because they show how

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well the market works in terms of matching supply and demand and how quickly customers can find what they want. For instance, if it takes an employer a long time to find an employee or a vendor to sell a product, this could signal that the marketplace is not operating efficiently and requires improvement.

Concentration or fragmentation of supply and demand
In simple terms, a market with a lot of different sellers and buyers is more valuable and sustainable. When there are many different people on both the supply and demand sides, it is less likely that one person leaving the market will significantly affect the market as a whole. Marketplaces can look at how much of the total business is done by the top sellers or buyers and try to spread the business out more evenly. One way to do this is to look at how much of a grocery delivery service's total sales come from each grocery store.

ECONOMICS-RELATED METRICS

Economics-related Metrics are the backbone of any successful business, giving you essential information about your company's financial health and potential for growth. From measuring income and costs to comprehending consumer behavior, these metrics provide businesses with a clear view of their current situation and future direction.

Pricing power
When a product or service becomes more valuable to its users, their willingness to pay increases. For instance, the value of a social network increases as its user base grows. Because of this, companies that use network effects can charge more for their products and services as they become more popular. Eventually, these businesses will be able to stop giving their goods away for free and start charging a lot

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more without seeing a drop in sales.

This is referred to as the "pricing power," and it is an essential indicator for firms to track in order to understand how their customers perceive their worth.

Unit economics

Businesses may provide fewer incentives to recruit new customers and improve their pricing power as the network's value to consumers grows. Businesses with local network effects, such as Thumbtack or Instacart, must monitor unit economics over time in each market to determine the relationship between market age, network density, and profitability. This can give valuable information about how to make the business more profitable and grow in a sustainable way.

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CASE STUDIES OF COMPANIES WITH NETWORKS EFFECTS

Have you ever wondered why some businesses seem unstoppable, growing quickly and taking over their fields? You can bet that they've taken advantage of network effects. We will look at some of the most successful companies that have used network effects to get to the top of their fields. From social media giant Facebook to ride-sharing king Uber, we'll look at the strategies they used to build and keep their vast networks going.

FACEBOOK

The success of Facebook can be traced to its grasp of network effects. At the time, the network was easier to use and less complicated than other social media platforms. Users invited their networks to try out this platform because it has a lot of value. As more and more users joined, the platform's value increased, making it more appealing for new users to join.

This self-reinforcing growth cycle contributed to Facebook's dominance in the social media sector. As new competitors came, Facebook's position didn't change because its network was extensive. Also, the company kept ahead of the curve by adding new features as they became popular. This helped it become the most popular social networking platform.

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AIRBNB

Airbnb is a prime example of a company that has utilized network effects to achieve astronomical growth and success. Airbnb developed a platform where supply and demand could readily meet by allowing individuals to rent out their homes or apartments to tourists. When more hosts joined the platform, travelers had more alternatives, which attracted more hosts. This virtuous growth cycle enabled Airbnb to become the world's largest accommodation provider, surpassing hotel chains in several areas. By building a good reputation and rating system, Airbnb earned the trust of its users, which strengthened its network effects and made it harder for competitors to copy its success.

MEDIUM

Medium is a platform for digital publishing that lets people write and share articles, stories, and other forms of written content. The platform is known for its simple and clean design, its focus on high-quality content, and its active community of users and writers.

Network effects have been a significant contributor to Medium's growth. As more people join the platform and post high-quality content, both viewers and authors will find it more useful. As a result, readers are more likely to join the platform because there is more content to read, and writers are more eager to join because their work has a larger audience.

Moreover, Medium has monetized its network effects by offering a membership service that gives users ad-free content access and other premium features. This lets Medium make money from its users as it gets more people to use it and creates network effects.

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Medium's success can be attributed to its ability to use network effects to create a valuable platform for readers and writers and turn this value into money through subscriptions and other revenue streams.

WHATSAPP

WhatsApp recognized the significance of network effects and utilized them to its advantage. They started out as a simple app for updating your status, but it quickly passed Facebook Messenger to become the most popular online chat service. This was done by giving users a free alternative to SMS messaging and eliminating the membership fee. This way of saving money attracted many users, who invited people from their networks to join the platform. As more people joined, the platform became more valuable, making it harder for other apps to copy and beat WhatsApp's success. The company was able to keep its market dominance because the network effect acted as a barrier.

UBER

Sharing economy platforms as Uber rely on network effects to achieve exponential growth. This is because when there are more customers, there are also more drivers, which means riders have shorter wait times and pay less. This attracts more riders, which in turn attracts more drivers, and so on. This creates a positive feedback loop that leads to a network effect in which the platform's value increases as more people use it. The company aims to build the best network in each area to make more money. Uber has surpassed the short-term risks and the most important challenge of building strong network effects. This gives them a competitive edge, especially now that their other divisions, like Uber Eats, are making money.

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WHEN NETWORK EFFECTS ARE NOT ENOUGH

It's important to know that a business can fail if it only depends on network effects to grow. Businesses often make two mistakes when they try to use network effects: they grow too quickly and in the wrong way. Too much growth can amplify business model faults, making them more difficult to solve. Airbnb, for instance, spent two years figuring out how to enable individuals to rent their homes to total strangers under conditions and for costs that satisfied both parties. The other error is improper expansion. This can occur when a firm caters to "power sellers," which can restrict the variety of things offered for sale and lower the average listing's quality.

In conclusion, network effects can be a strong instrument for business growth. Still, it is crucial to be aware of the possible hazards and to grow in a manner that is both sustainable and beneficial to the firm and its customers.

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HOW TO BEAT NEGATIVE NETWORK EFFECTS

Network effects can be double-edged in the business world. On the one hand, they can lead to growth that grows faster and faster, creating a cycle of success that keeps going and going. On the other hand, negative network effects could lead to a downward spiral of failure. The key to success is knowing how to take advantage of network effects while avoiding problems.

There are ways to get around negative network effects and change the way things are going. It is possible to overcome the odds and emerge successful through targeted marketing initiatives, collaborations with enterprises, or other strategies.

EMBEDDED CURATION

Embedded curation is a way to control how people interact on a network by putting limits on what they can do with a product. This kind of curation is built into the limits of the product, which makes it a flexible method that can be used on any kind of network. This is illustrated by Nextdoor, which restricts user interaction to surrounding communities, reducing network congestion in comparison to a more general network such as Facebook. Because of this, specialized or "vertical" networks often give customers a better experience than "horizontal" networks. Also, data networks like Mapbox that use passive crowdsourcing are less likely to have bad network effects because the product

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controls what users share.

USER-CONTROLLED CURATION

To combat negative effects, businesses might give users greater control over their platform interactions. This can be accomplished by giving users tools such as filters and prioritized lists that allow them to pick with whom they communicate and what information they view. On marketplace sites like Deliveroo or Uber Eats, for example, users can filter their search results to only show restaurants that match their tastes. In the same way, Airbnb users can filter their search results so that they only see places to stay that fit their budget and location needs. Users of social media platforms such as Facebook can create friend lists to control who they disclose information. These filters and curation tools can help users find relevant relationships and content, which increases the positive network effects of the platform. It is also crucial to note that too many filters can be overwhelming for customers and should be used cautiously.

MANUAL CURATION

Manual curation is a method for a corporation or network owner to actively manage network interactions and participants. This may involve enforcing rules and deleting content, supplies, or participants who violate them. The main benefit of this method is that it can have an effect right away and is effective against bad users. However, it is less effective against network pollution and does not strengthen beneficial contacts. Also, manual curation is hard to use on a large scale and works best for short-term solutions in the early stages of a startup's growth. It applies to all network kinds, including social networks, marketplaces, platforms, and data networks.

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COMMUNITY CURATION

Community curation enables users to promote or demote information or supply units on a network without the product owner's involvement. This may include ratings, likes, shares, etc., depending on the specific network. Community curation's primary advantage is that it is scalable, can create beneficial network effects, and can, to some extent, minimize network pollution. Nonetheless, it requires an active network and is less effective against bad actors. So, it works best for networks that have already reached the right level of liquidity and have a system in place for manual curation. It is important to remember that community curation can cause problems if it is used by itself, especially early in the adoption cycle.

AUTOMATED CURATION

Automated curation is a way for networks to try and find a balance between controlling interactions and allowing users to promote or demote content or supply units on their own. It involves using data from user behavior and engagement, performance metrics, and community curation signals like likes and ratings to create a ranking algorithm. It is most effective when implemented after a network has reached a certain level of liquidity, as there is enough information to guide the curation. The news feed algorithm on Facebook, the Superhost program on Airbnb, and the restaurant ranking algorithm on Uber Eats are all examples of automated curation. It can be used in any network where people actively look for and share information. This includes two-sided and multi-sided marketplaces, platforms, and even some one-sided networks.

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CONCLUSION

In conclusion, network effects are a powerful weapon in the business world, and organizations that understand how to utilize them can reap substantial benefits. Businesses like Facebook and WhatsApp have done very well by using network effects to attract more users.

However, network effects also have potential drawbacks, such as negative network effects, which can harm a corporation if not managed effectively. Firms can deal with these problems in a number of ways, including user-controlled curation, manual curation, curation by the community, and automated curation. Each of these approaches has its own benefits and drawbacks, and the most effective strategy will depend on the particulars of the organization. In the modern economy, businesses that want to grow and succeed must understand and be able to control network effects.

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ABOUT THE AUTHOR

Daniel Pereira is a Brazilian-Canadian entrepreneur that has been designing and analyzing business models for over 15 years. You can read more about his journey as a Business Model Analyst **here**.



E-mail Daniel if you have any questions at: daniel@businessmodelanalyst.com
You can connect with Daniel at Linkedin: https://www.linkedin.com/in/dpereirabr/

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